

How to succeed

- Read the Chapter ahead of lectures
- Come to class
- Start paper early
- Study groups
- Prepare exams
- Don't cheat, plagiarize, or otherwise participate in un-ethical behavior
- **Learn with Internet: Wikipedia, google(baidu)...**
- Ask questions and Think skeptically

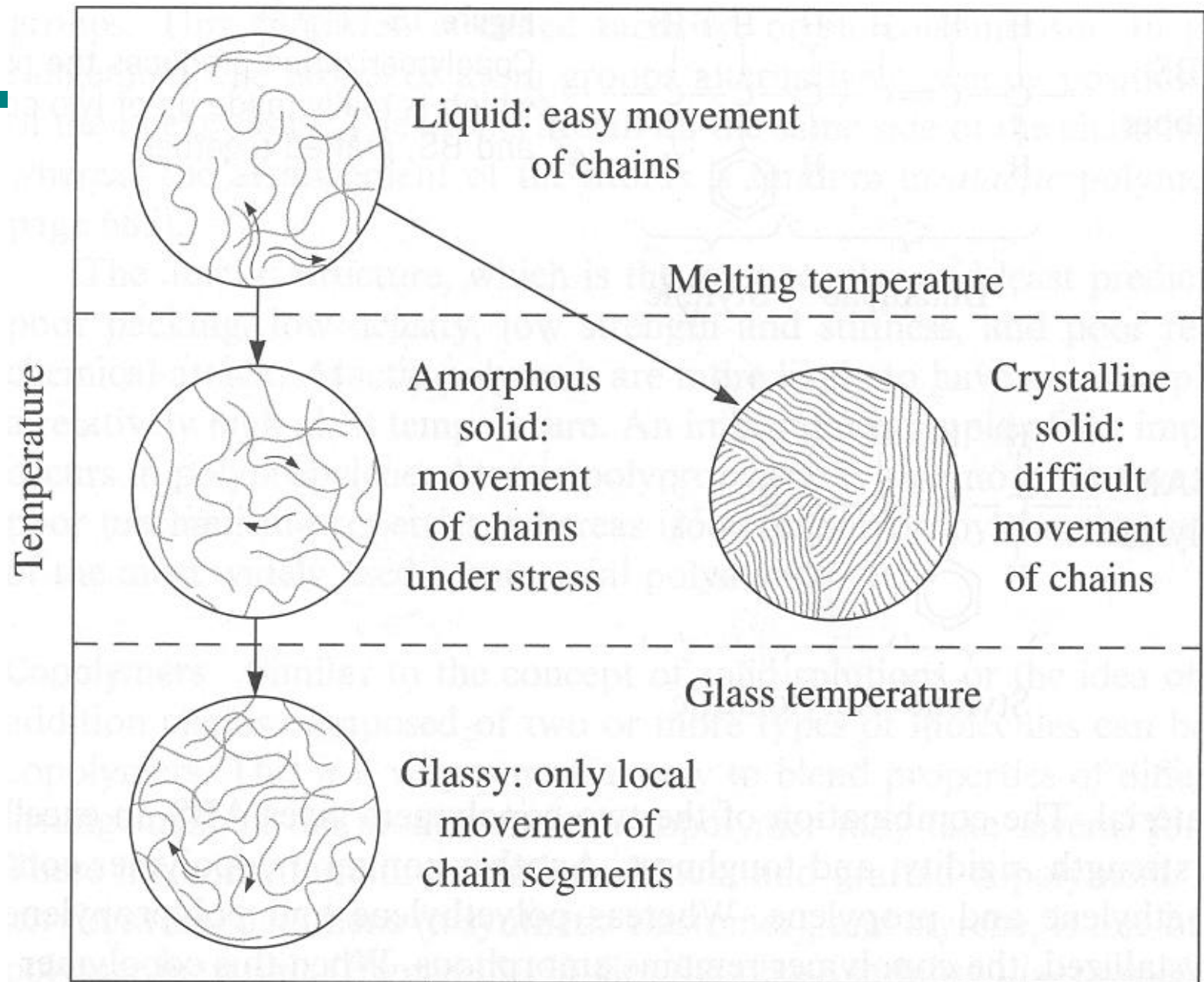
Important Points from Last Lecture

- The root-mean-squared end-to-end distance, $\langle R^2 \rangle^{1/2}$, of a freely-jointed polymer molecule is $N^{1/2}a$, when there are N repeat units, each of length a .
- The radius-of-gyration of a polymer, R_g , is $(1/6)^{1/2}$ of its root-mean-square end-to-end distance $\langle R^2 \rangle^{1/2}$.
- Expansion factor
- Theta temperature
- χ parameter
- Solubility parameter

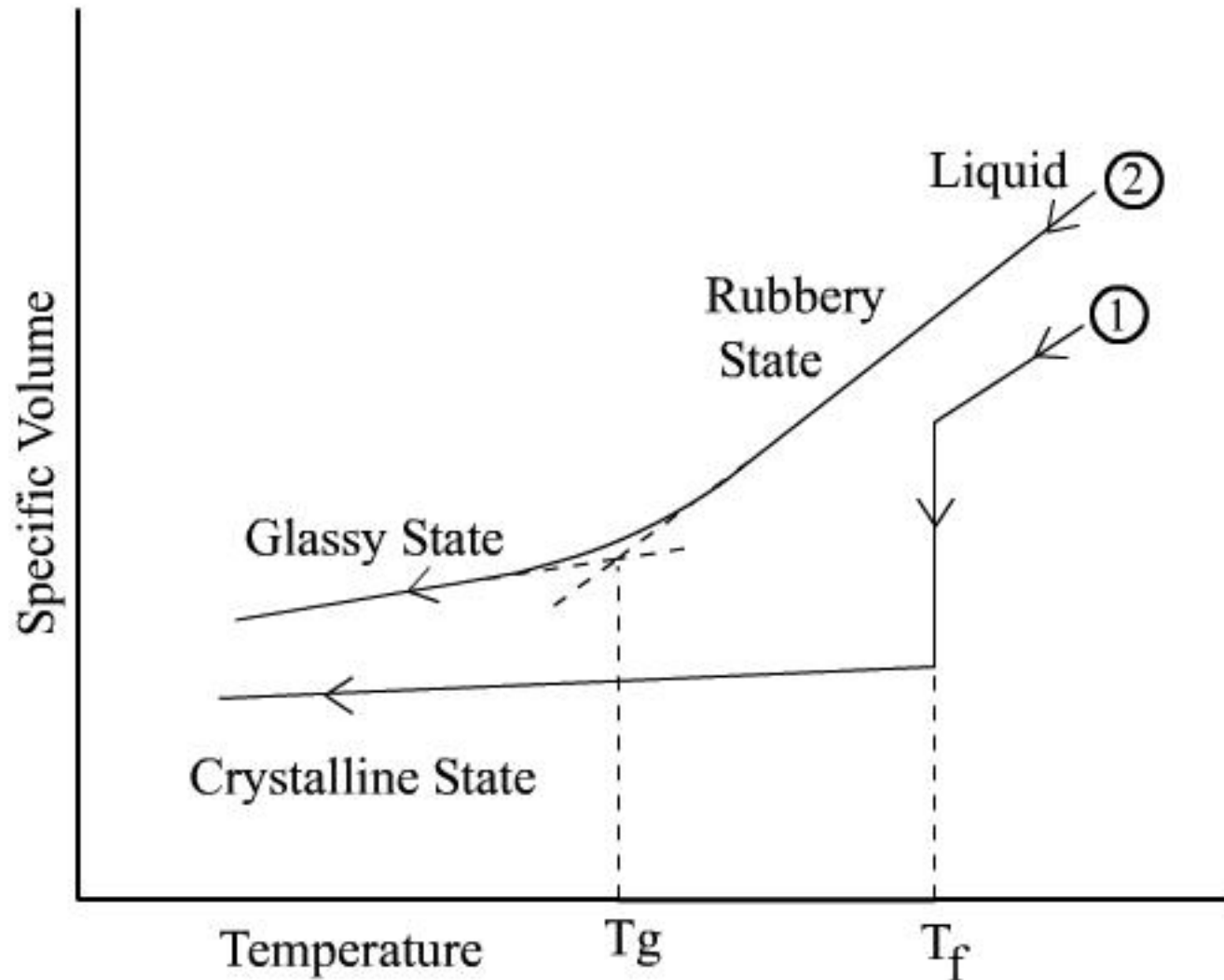
Introduction to Polymer Materials

Lecture 4-5: Solid State Properties I

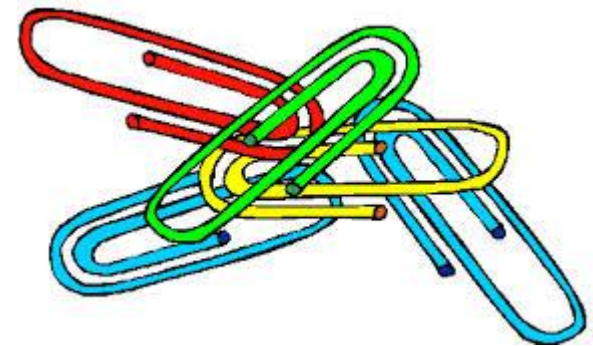
SUN Dazhi



Glass transition temperature



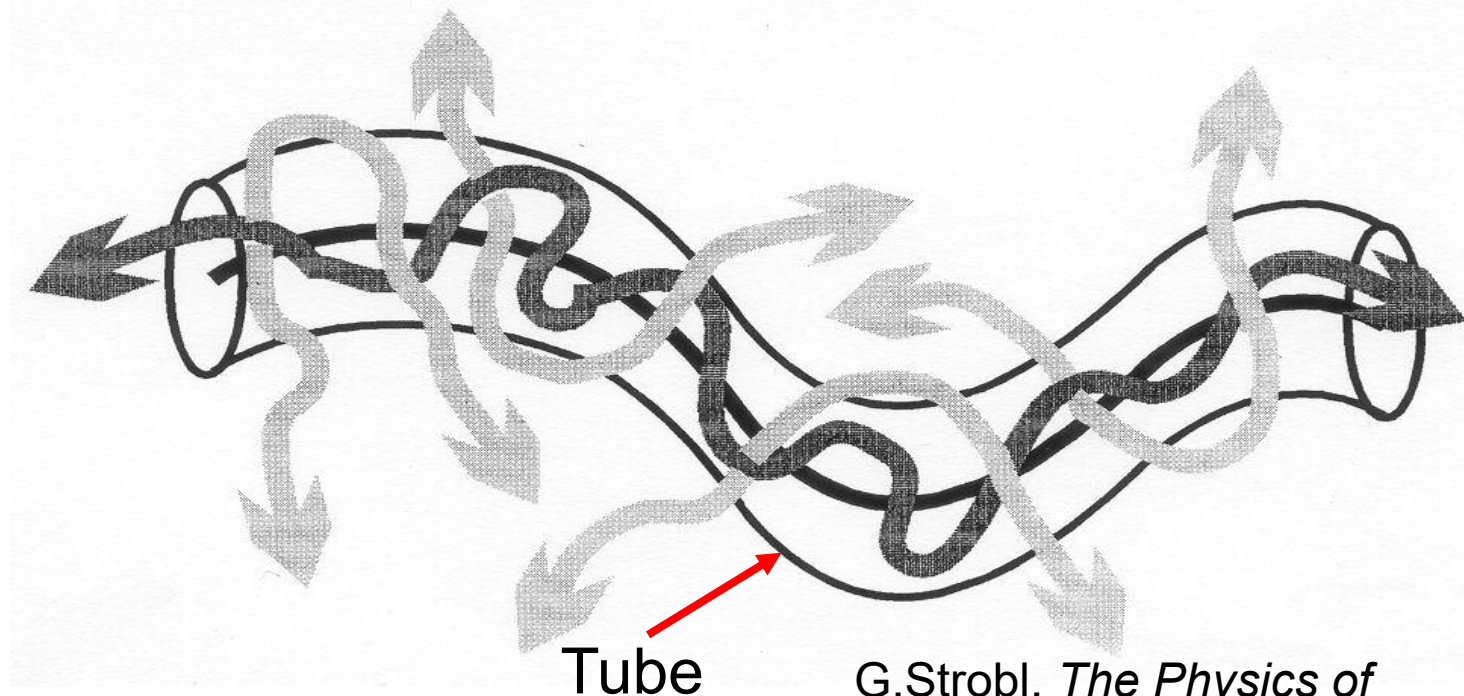
Polymer Chain Entanglements



Concept of Chain Entanglements

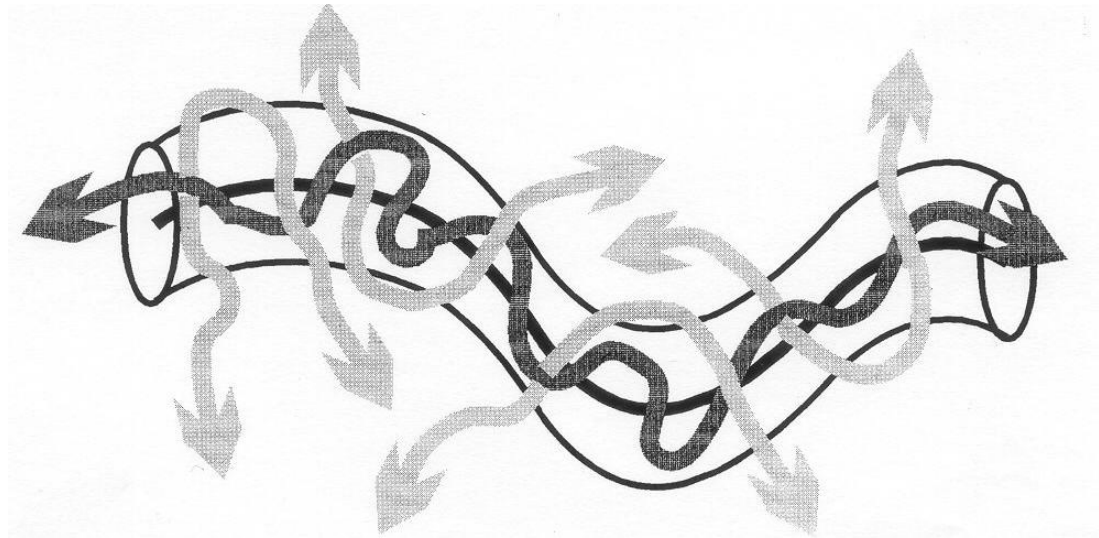
If the molecules are sufficient long ($N > 100$ - corresponding to the **entanglement mol. wt., M_e**), they will entangle with each other.

Each molecule is confined within a dynamic “tube”.



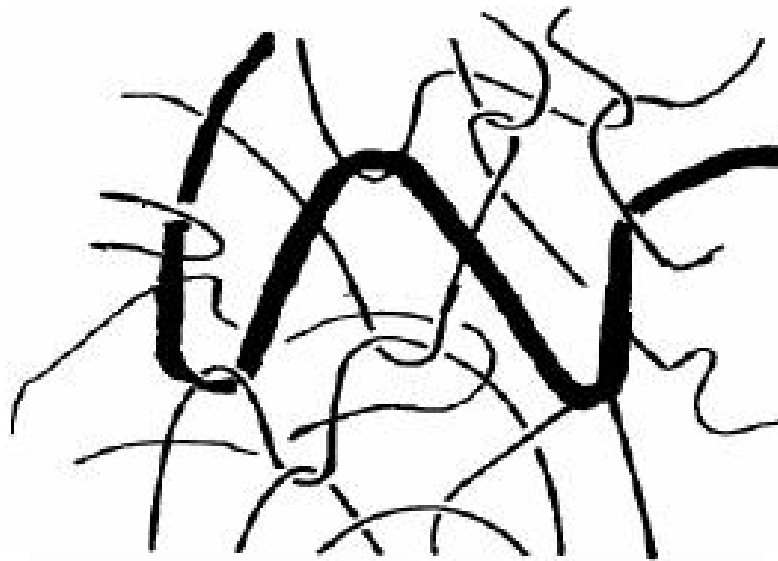
G. Strobl, *The Physics of Polymers*, p. 283

Network of Chain Entanglements



There is a direct analogy between chemical crosslinks in rubbers and “physical” crosslinks that are created by the entanglements.

The physical entanglements can support stress (for short periods up to a time τ_T), creating a “transient” network.



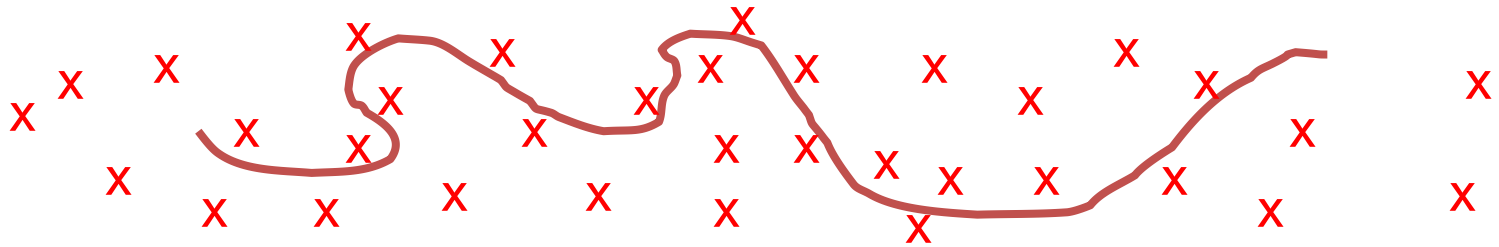
*Reptating chain
entangled*

An Analogy



- There are obvious similarities between a collection of snakes and the entangled polymer chains in a melt.
- The source of continual motion on the molecular level is thermal energy, of course.

Reptation Theory



- Polymer molecules “dis-entangle” after a time, t_{Tube} .
- The process by which a polymer chain moves through its tube formed by entanglements is called “reptation”.
- Reptation (from the Latin *reptare*: “to crawl”) is a snake-like diffusive motion that is driven by thermal motion.

Reptation Scaling Theory

Pierre de Gennes (Paris) developed the concept of polymer reptation and derived scaling relationships.



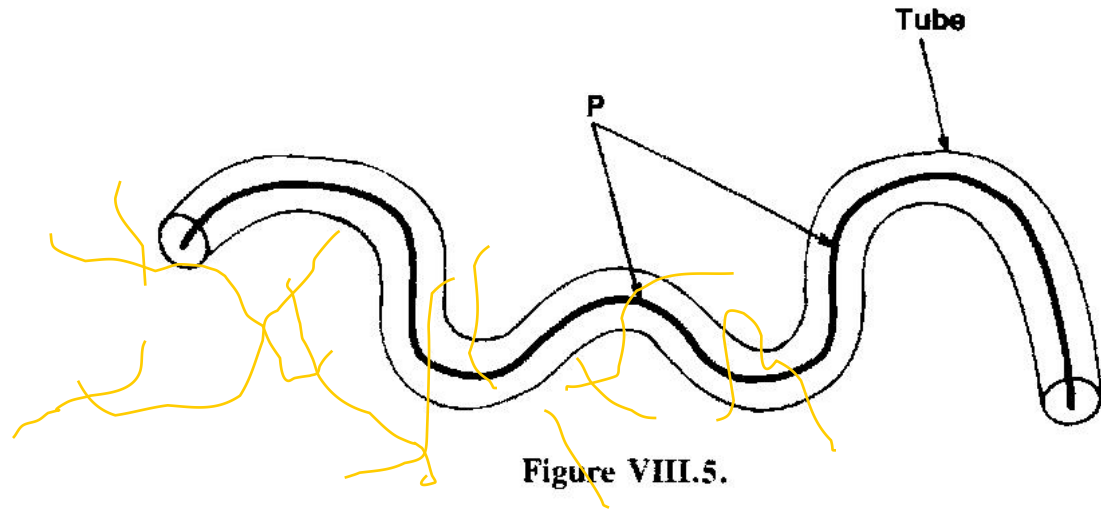
In 1991, de Gennes was awarded the Nobel Prize for Physics.

Reptation Scaling Theory

“They’re like snakes”

“As they follow a curve, the large ones can hardly advance.”

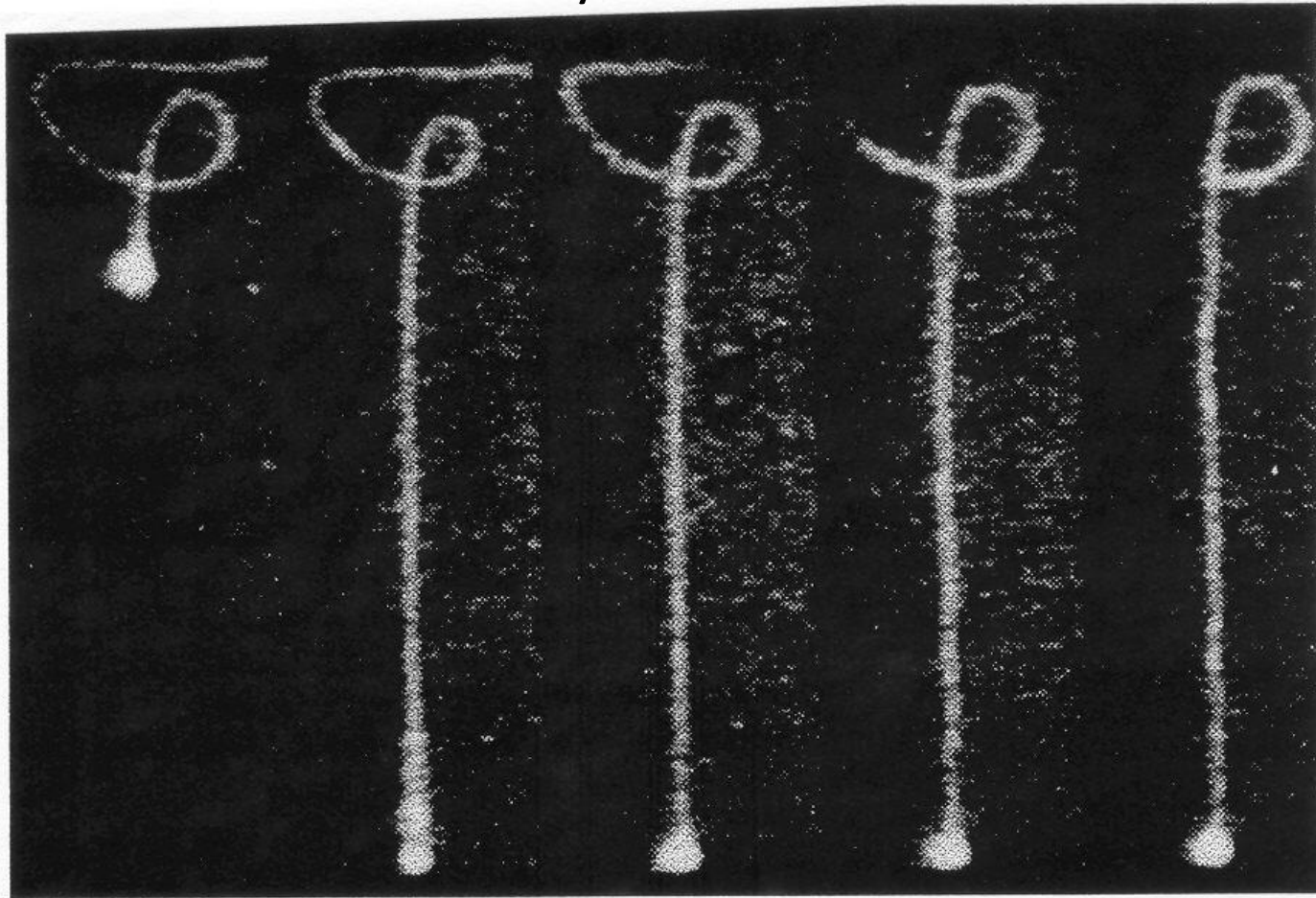
$$\tau_{tube} \sim N^3$$



P.G. de Gennes
Scaling Concepts in Polymer Physics
Cornell University Press, 1979

Experimental Evidence for Reptation

Fluorescently-stained DNA molecule



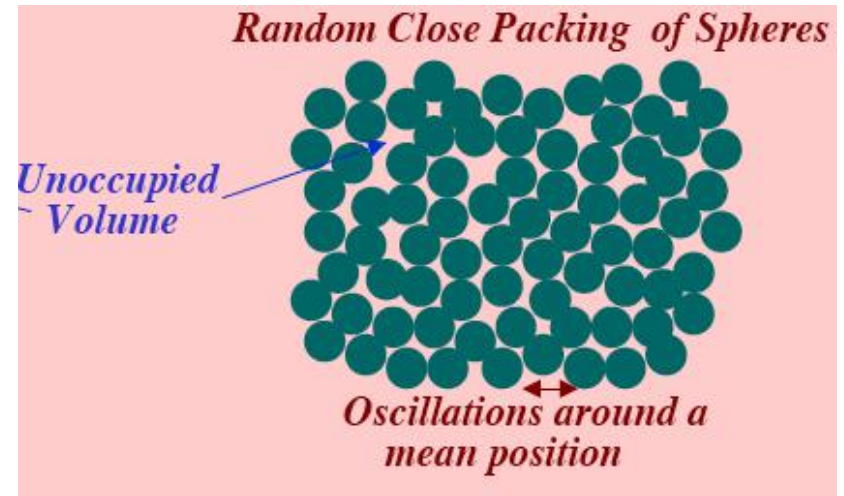
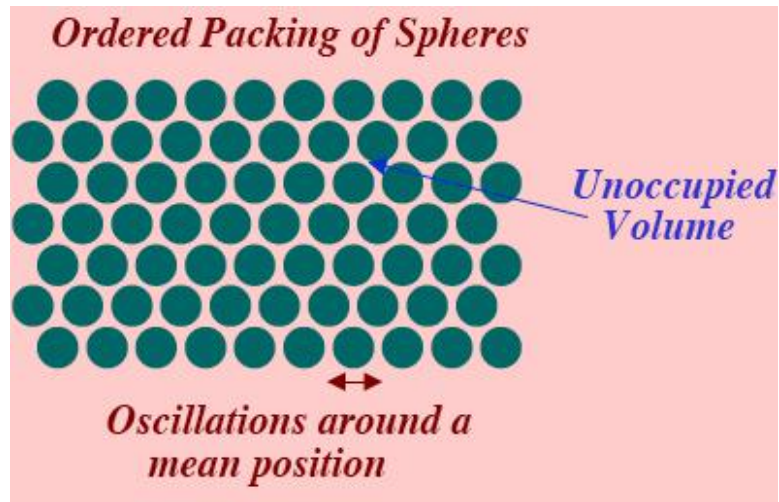
Initial state

Stretched

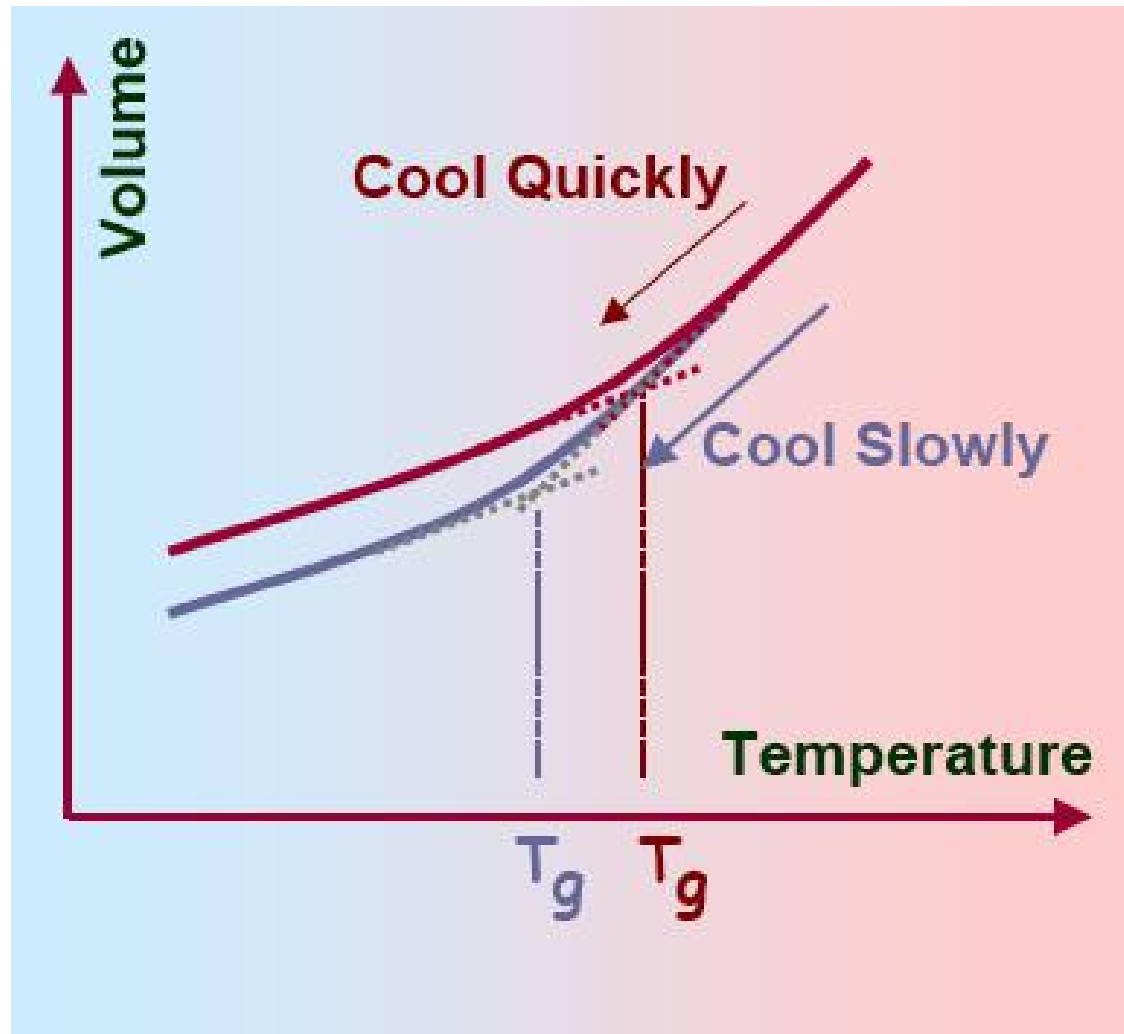
Chain follows the path of the front

Chu et al., *Science* (1994) **264**, p. 819.

Amorphous Glasses ($< T_g$)



Rate of cooling affects T_g



Polymer

T_g (° C)

Polyacenaphthalene	264
Polyvinyl pyrrolidone	175
Poly- <i>o</i> -vinyl benzyl alcohol	160
Poly- <i>p</i> -vinyl benzyl alcohol	140
Polymethacrylonitrile	120
Polyacrylic acid	106
Polymethyl methacrylate	105
Polyvinyl formal	105
Polystyrene	100
Polyacrylonitrile	96
Polyvinyl chloride	87
Polyvinyl alcohol	85
Polyvinyl acetal	82
Polyvinyl propional	72
Polyethylene terephthalate	69
Polyvinyl isobutyral	56
Polycaprolactam (nylon 6)	50
Polyhexamethylene adipamide (nylon 6,6)	50
Polyvinyl butyral	49
Polychlorotrifluoroethylene	45
Ethyl cellulose	43
— Polyhexamethylene sebacamide (nylon 6,10)	40
Polyvinyl acetate	29

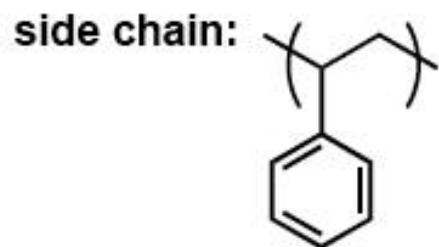
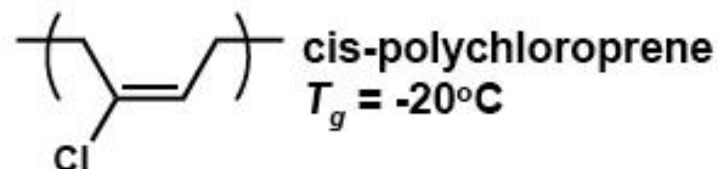
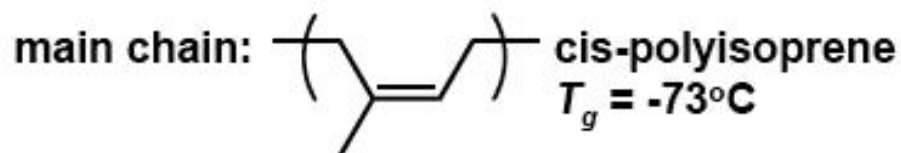
Polymer

T_g (° C)

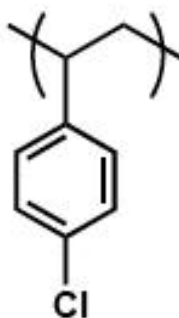
Polyperfluoropropylene	11
Polymethyl acrylate	9
Polyvinylidene chloride	-17
Polyvinyl fluoride	-20
Poly-1-butene	-25
Polyvinylidene fluoride	-39
Poly-1-hexene	-50
Polychloroprene	-50
Polyvinyl- <i>n</i> -butyl ether	-52
Polytetramethylene sebacate	-57
Polybutylene oxide	-60
Polypropylene oxide	-60
Poly-1-octene	-65
Polyethylene adipate	-70
Polyisobutylene	-70
Natural rubber	-72
Polyisoprene	-73
Polybutadiene	-85
Polydimethyl siloxane	-123

Factors that affect T_g

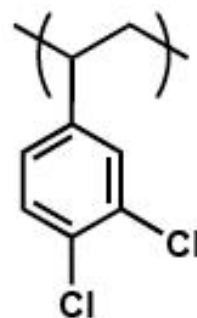
- Polar groups increase packing density;
- more thermal energy is needed to create volume



$T_g = 100^\circ\text{C}$



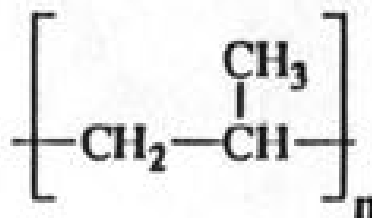
$T_g = 110^\circ\text{C}$



$T_g = 128^\circ\text{C}$

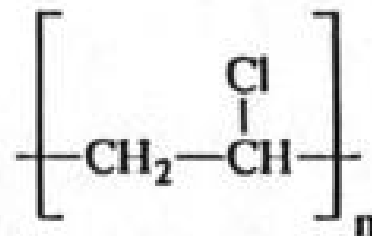
Factors that affect T_g

Atactic
Polypropylene



$$T_g = -10^\circ\text{C}$$

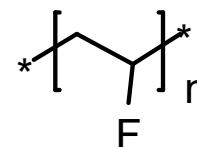
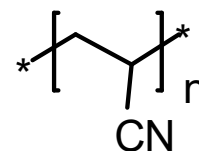
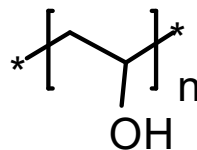
Atactic
Poly(vinyl chloride)



$$T_g = 87^\circ\text{C}$$

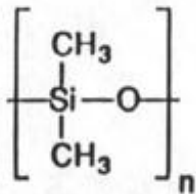
- * The chlorine atom and methyl group have approximately the same effect on bond rotations.
- * The polar character of the Cl atom leads to stronger forces of attraction between chains, however, which means that on average those groups engaging in such interactions are closer together.
- * The free volume is less and the T_g is higher (reality is probably a bit more complicated than this, but this simple explanation gives a good feel for the major effect).

Other polar vinyl polymer:



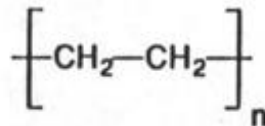
Factors that affect T_g

- * Chain stiffness affects chain mobility.
- * If there are bulky groups, such as benzene rings, in the backbone of the polymer chain, there is a high energy barrier to rotations, which then only occur at higher temperatures.



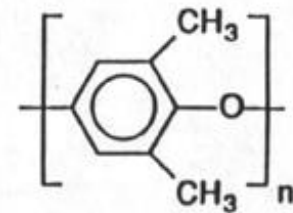
Poly(dimethyl siloxane)

$$T_g \cong -120^\circ\text{C}$$



Polyethylene

$$T_g \cong -80^\circ\text{C}$$



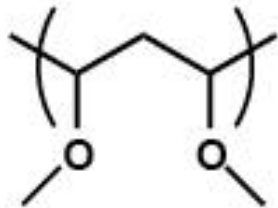
Poly(phenylene oxide)

$$T_g \cong +200^\circ\text{C}$$

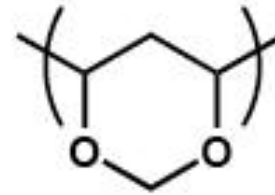
Increasing Chain Stiffness



Factors that affect T_g



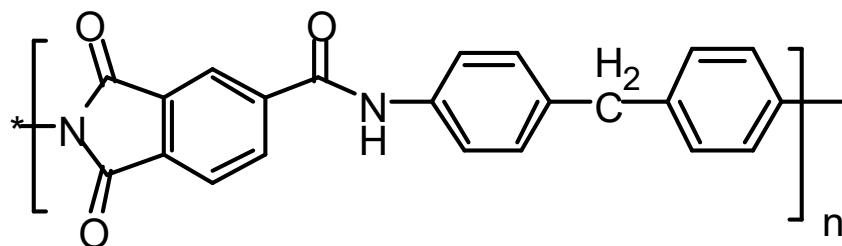
poly(vinyl methyl ether)
 $T_g = -13^\circ\text{C}$



poly(vinylformal)
 $T_g = 105^\circ\text{C}$

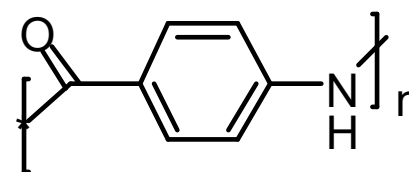
Main chain stiffness: reduced flexibility

High Tg Polymers



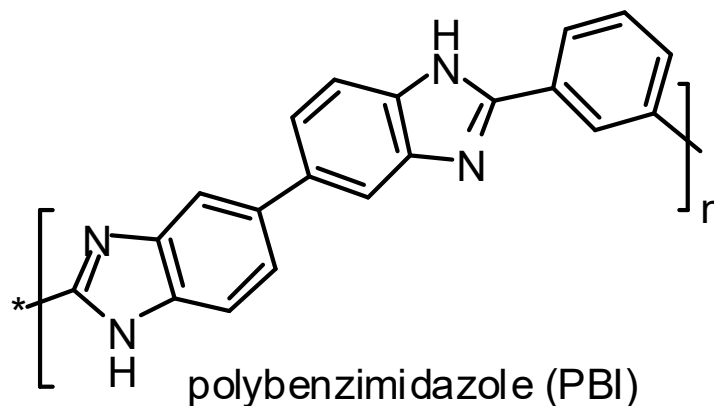
polyamide imide (Torlon)

$$T_g = 275\text{ }^{\circ}\text{C}$$



polybenzamide

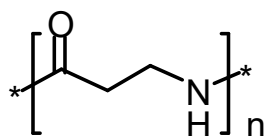
$$T_g = 327\text{ }^{\circ}\text{C}$$



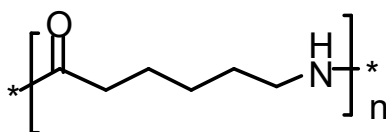
polybenzimidazole (PBI)

$$T_g = 425\text{ }^{\circ}\text{C}$$

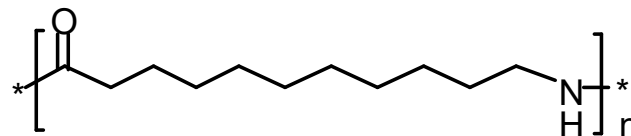
Nylons or polyamides



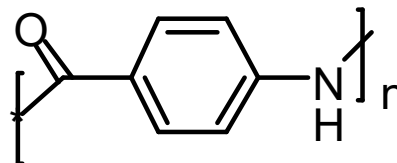
Nylon-3
 $T_g = 110\text{-}200\text{ }^{\circ}\text{C}$



Nylon-6
 $T_g = 52\text{ }^{\circ}\text{C}$



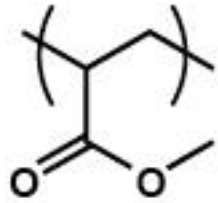
Nylon-11
 $T_g = 42\text{ }^{\circ}\text{C}$



polybenzamide

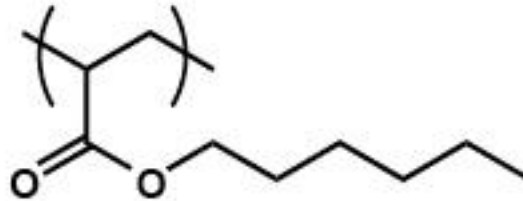
$$T_g = 327\text{ }^{\circ}\text{C}$$

Factors that affect T_g



Side Chain Rigidity

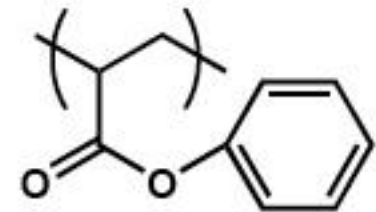
Poly(methyl acrylate)
 $T_g = 10^\circ\text{C}$



Long chains plasticize

Poly(n-hexyl acrylate)
 $T_g = -57^\circ\text{C}$

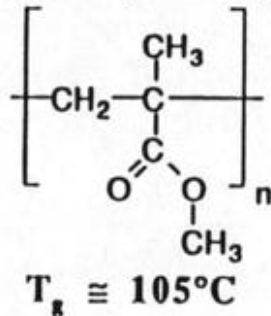
Anchors to movement



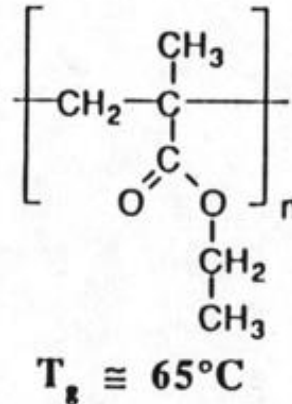
Poly(phenyl acrylate)
 $T_g = 57^\circ\text{C}$

Long chains plasticize movements

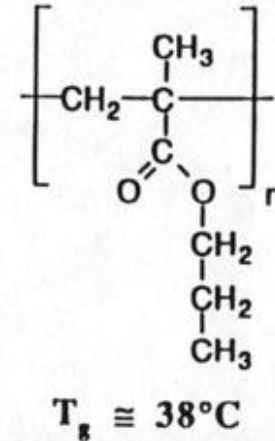
Poly(methyl methacrylate)



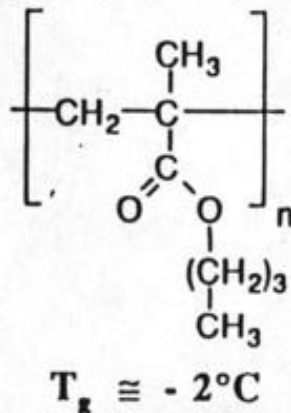
Poly(ethyl methacrylate)



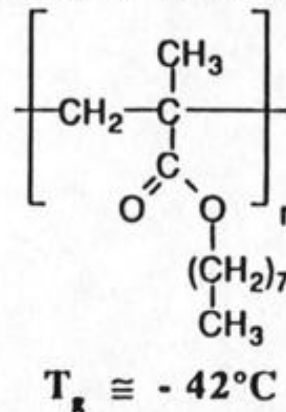
Poly(propyl methacrylate)



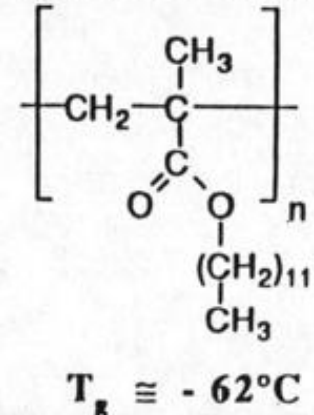
Poly(butyl methacrylate)



Poly(octyl methacrylate)

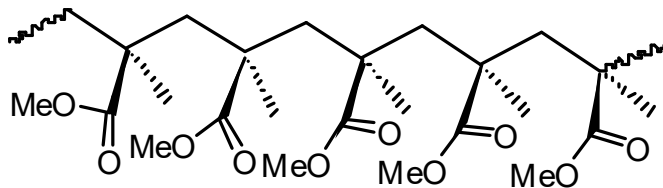


Poly(dodecyl methacrylate)

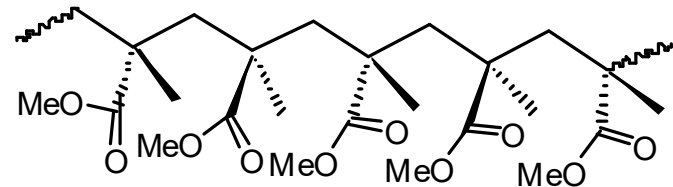


Factors that affect T_g

poly(methyl methacrylate)

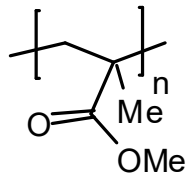


$T_g = 47\text{ }^{\circ}\text{C}$ (isotactic)



$T_g = 120\text{-}140\text{ }^{\circ}\text{C}$ (syndiotactic)

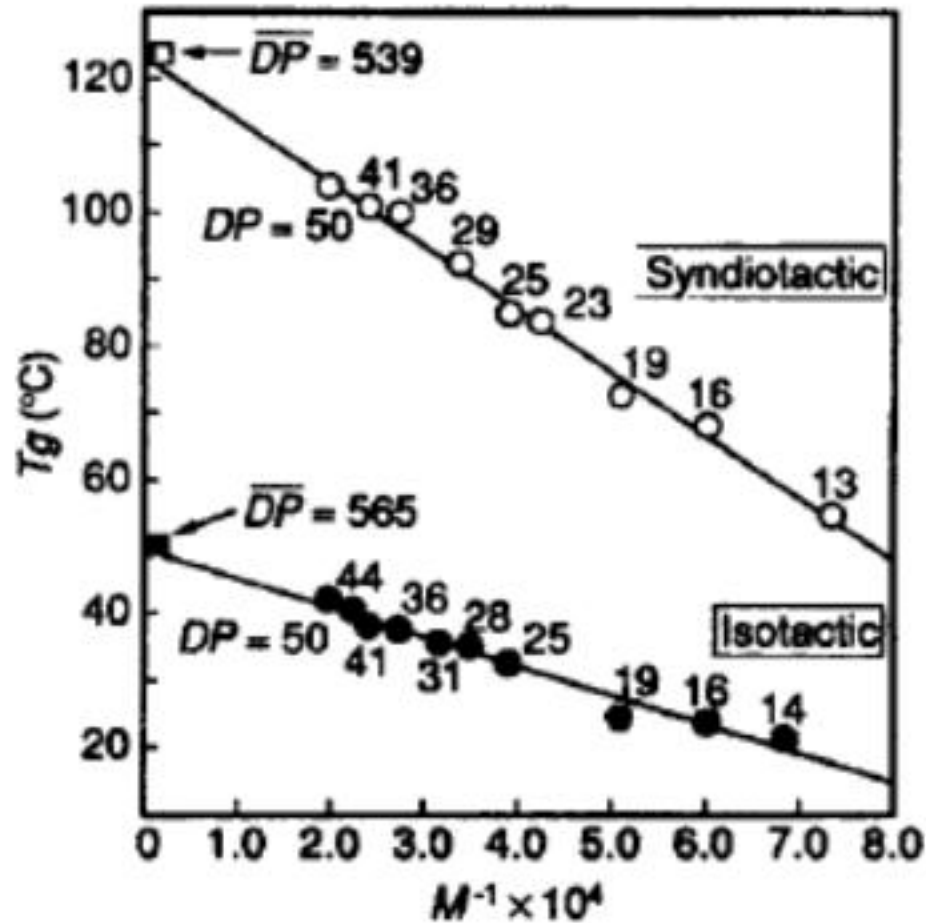
poly(methyl methacrylate)



$T_g = 110\text{ }^{\circ}\text{C}$ (atactic > 50 % syndiotactic)

Factors that affect Tg

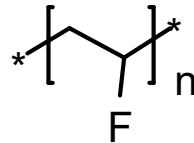
Tacticity



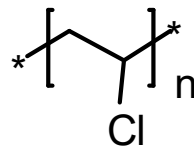
Factors that affect T_g

Symmetry of substituents

asymmetric

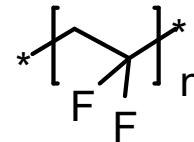


$$T_g = -20\text{ }^{\circ}\text{C}$$

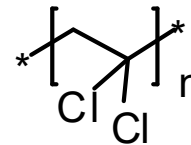


$$T_g = 87\text{ }^{\circ}\text{C}$$

symmetric



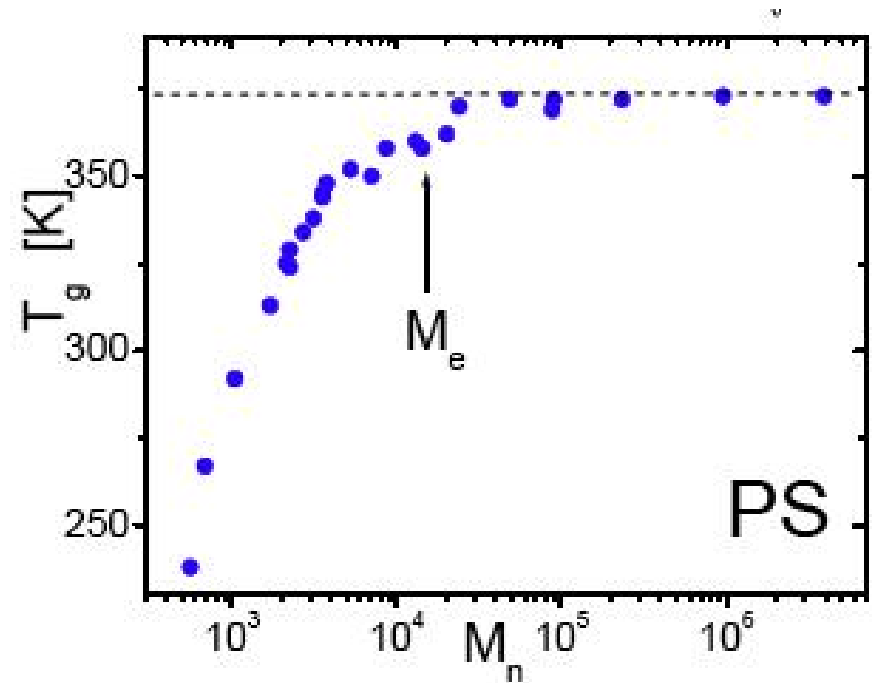
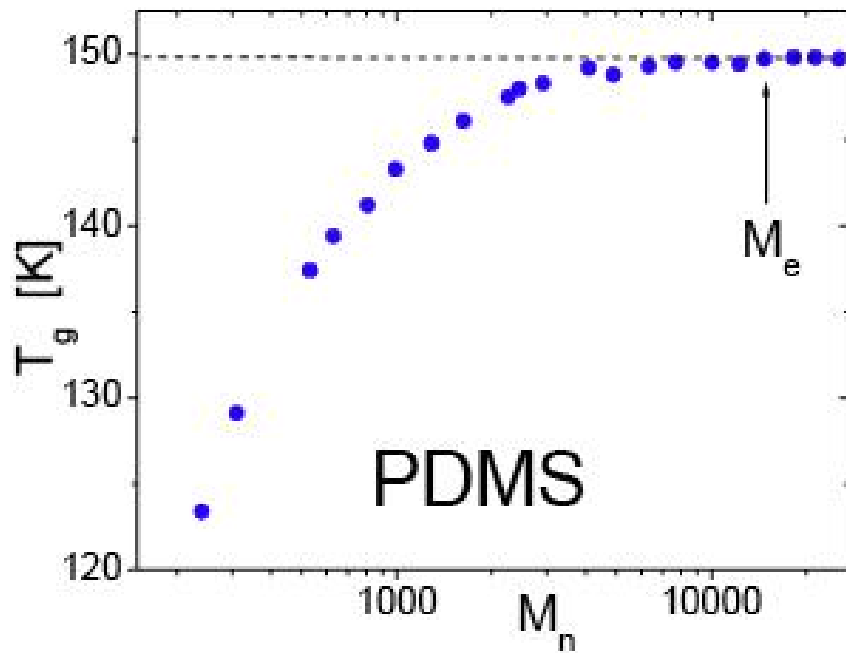
$$T_g = -39\text{ }^{\circ}\text{C}$$



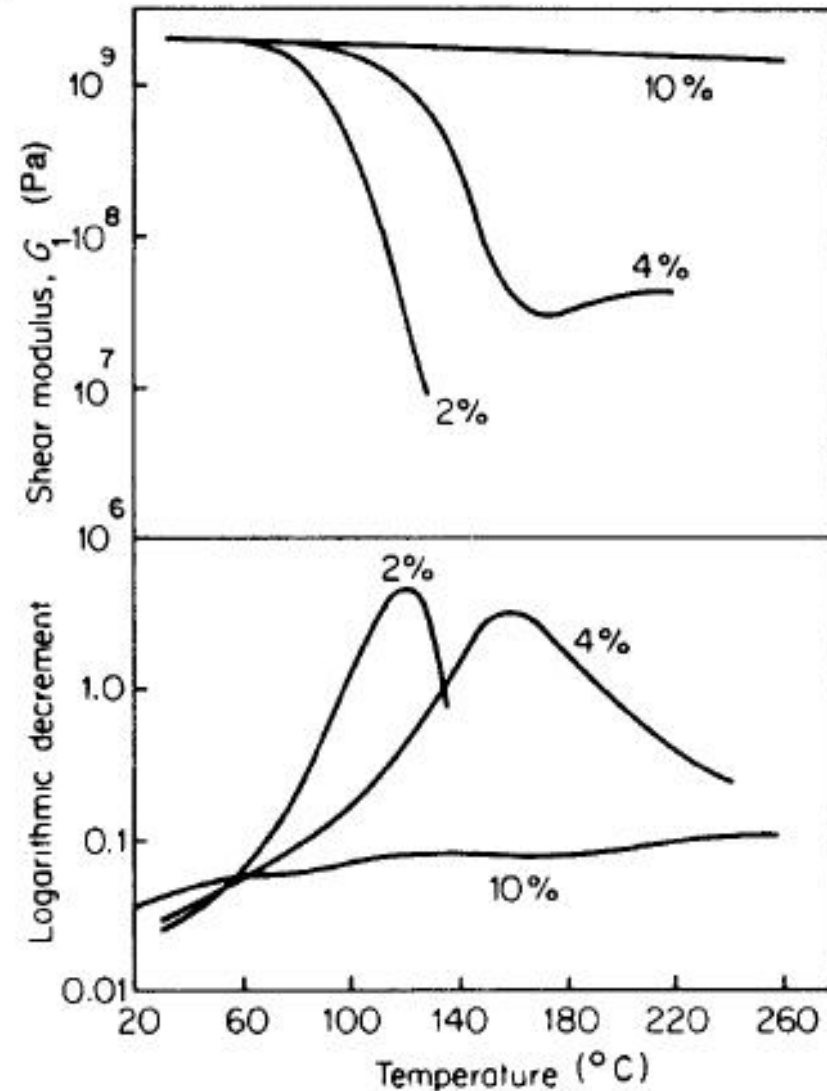
$$T_g = -17\text{ }^{\circ}\text{C}$$

Asymmetric have higher T_g's

Factors that affect Tg: Mw

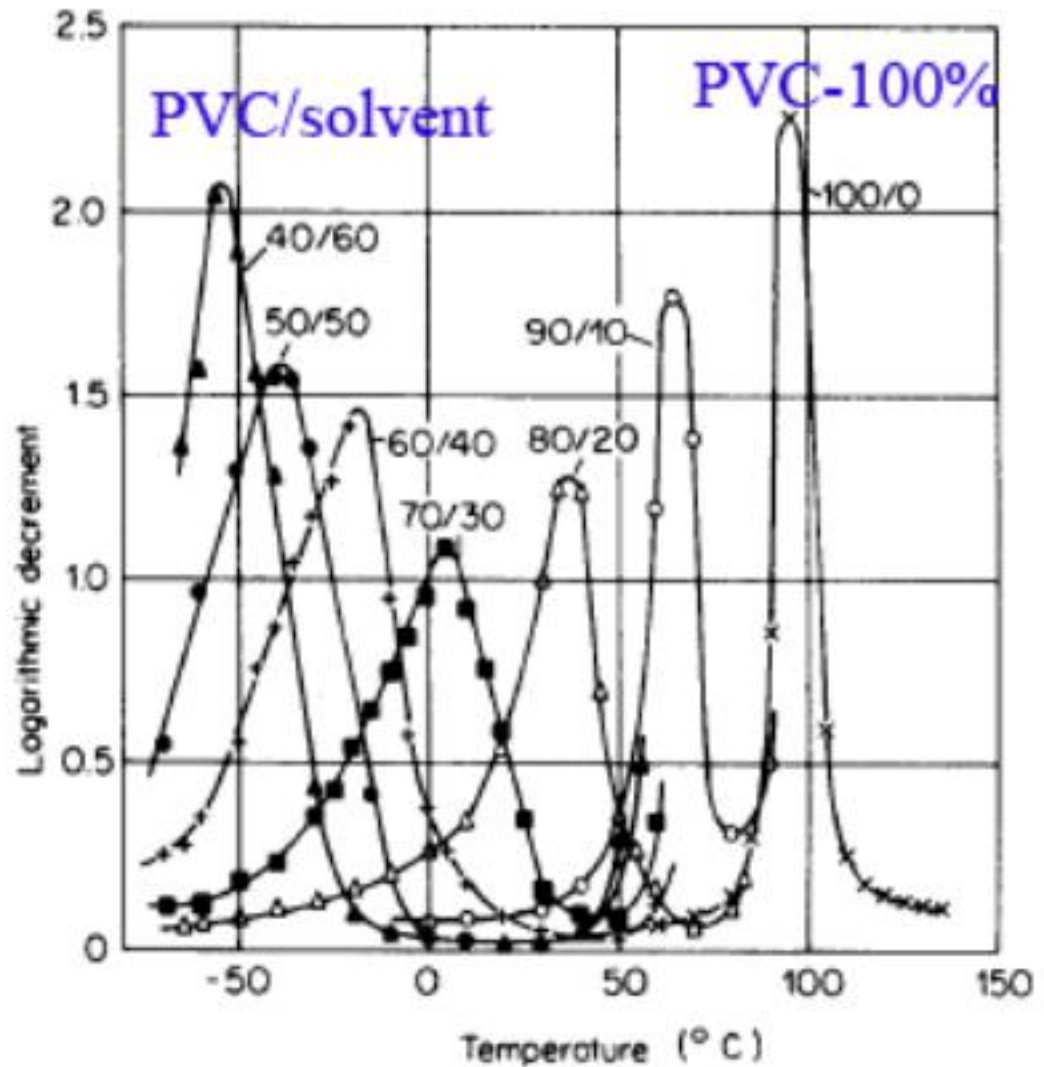
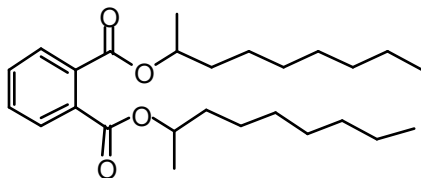


Factors that affect Tg: Crosslinking

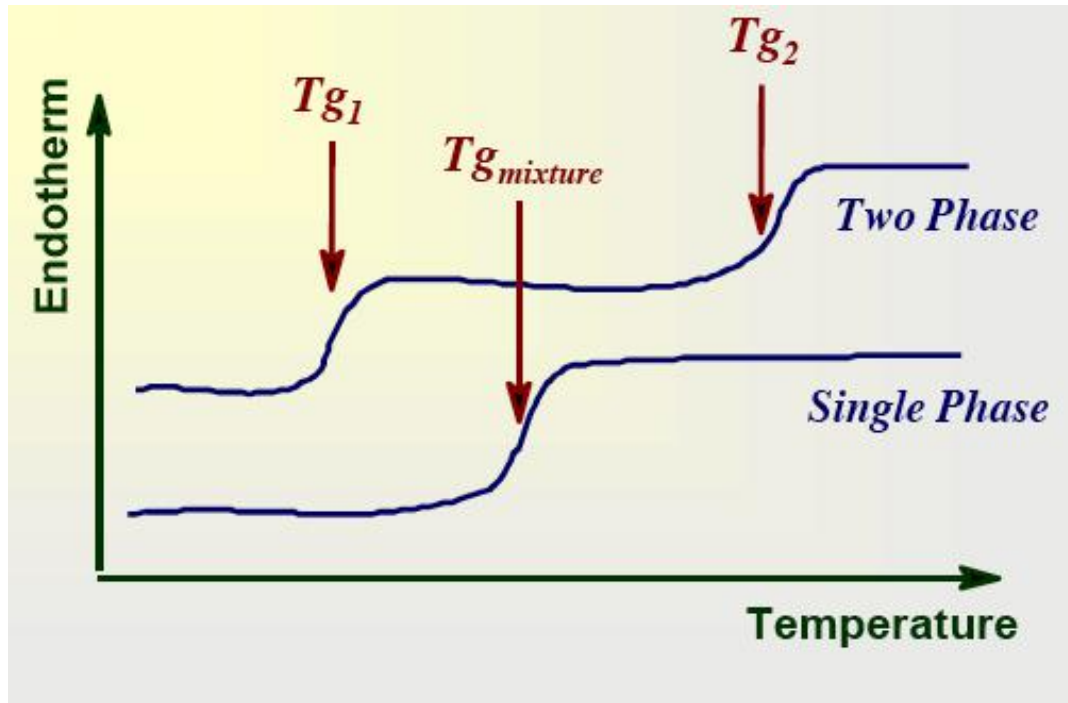


Factors that affect Tg: Plasticizer

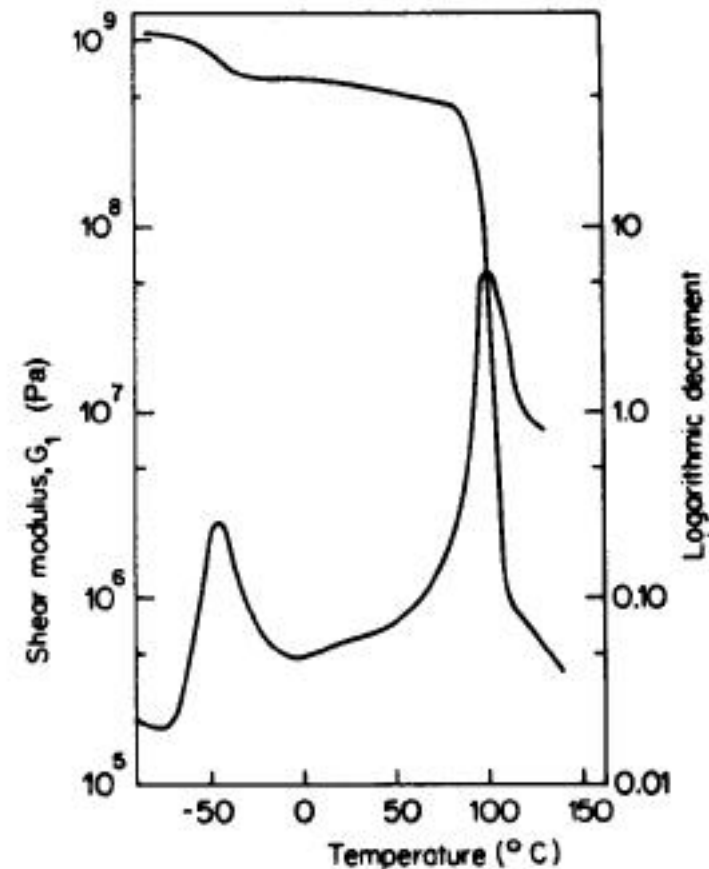
Phthalates



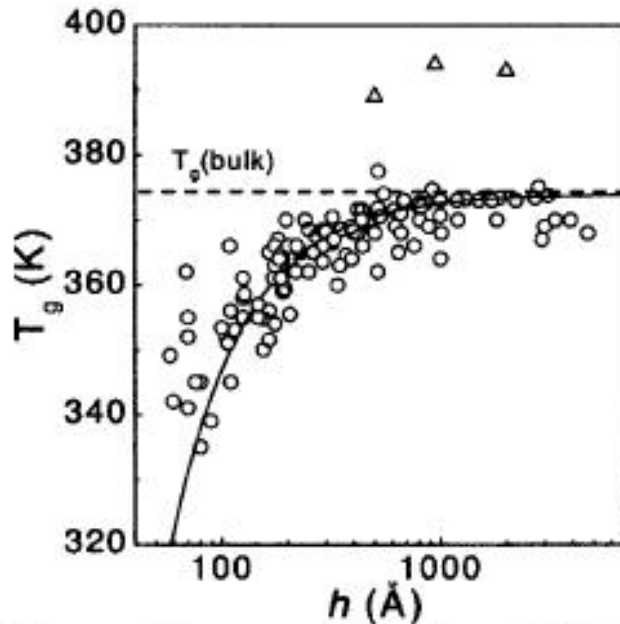
Immiscible (Two phase) and miscible (blends) polymers



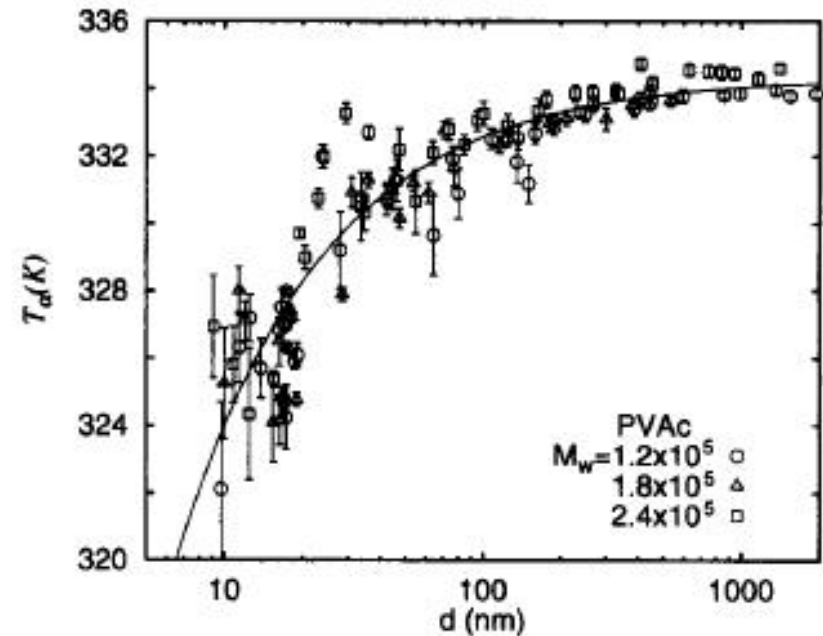
Schematic diagram depicting the intermediate T_g observed in
(i) miscible blends of two polymers (eg PVC and PCL)
(ii) a polymer and a plasticizer (eg PVC and dioctyl phthalate)
(iii) random copolymers (eg styrene-co-butadiene SBR)



T_g as a function of film thickness



T_g in thin PS films as a function of thickness h .

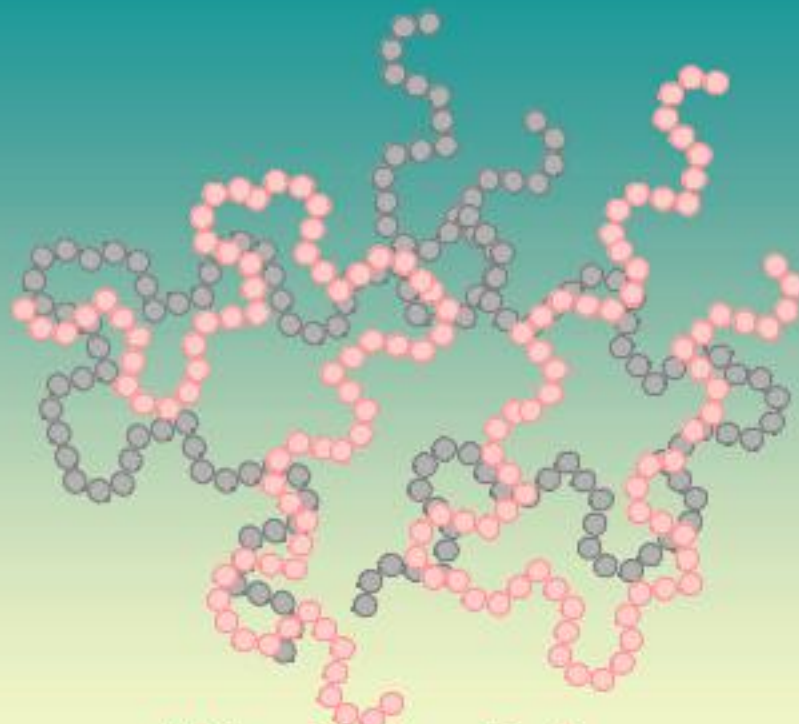
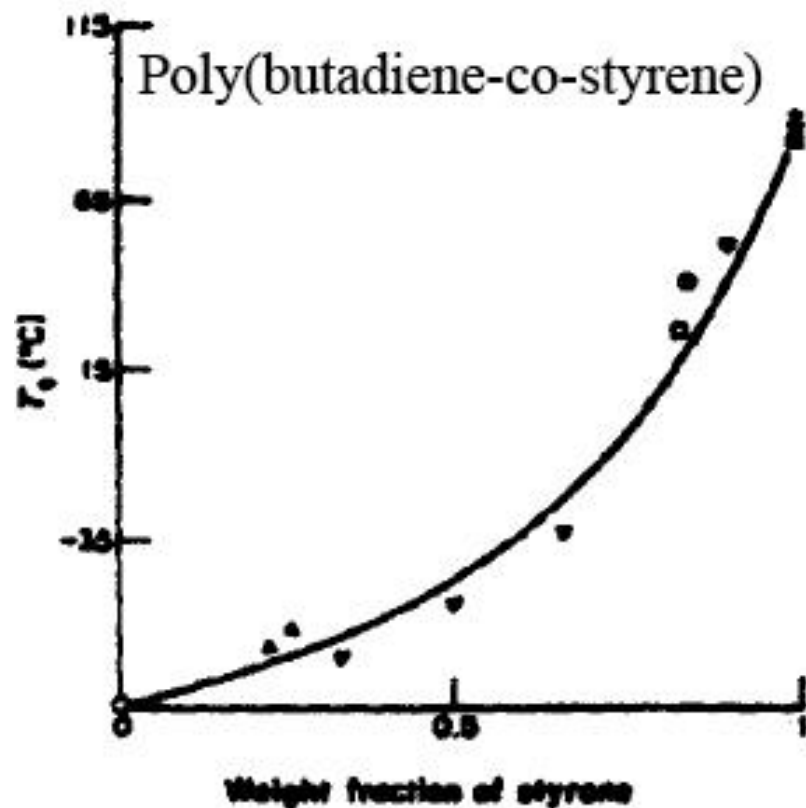


T_g in thin PVAc films [Fukao, et al., *J.Non.Cryst.Sol.* 307-310, 517 (2002)].



Random Copolymer

$$\frac{1}{T_g} = \frac{W_1}{T_{g_1}} + \frac{W_2}{T_{g_2}}$$



Miscible Polymer Blend

Glass Transition Temperature

- Rigid group in backbone
- Flexible polymer backbone
- Steric Hinderance
- Long plasticizing side groups
- Symmetrical substituents
- Polar functionalities
- Plasticizers



HOMework

- Describe at least **3** methods to determine the T_g of polymers.